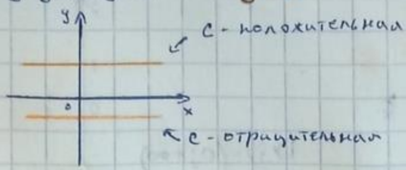
Кочеткова Мария Павловна 2.2

Основные (или простейшие) элементарные ф-ции:

• постоянная функция — $y = C$

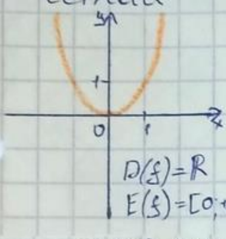
$D(f) = \mathbb{R}$

$E(f) = \{c\}$



• степенная функция — $y = x^{\alpha}$, $\alpha \in \mathbb{R}$

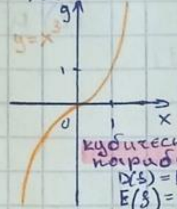
(1) α - целая четная



$D(f) = \mathbb{R}$

$E(f) = [0; +\infty)$

(2) α - целая нечетная

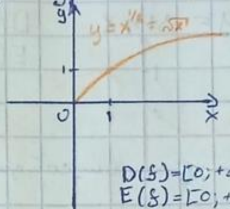


кубическая парабола

$D(f) = \mathbb{R}$

$E(f) = \mathbb{R}$

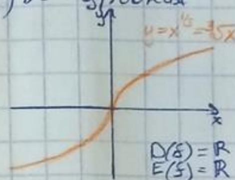
(3) α - четная дробная



$D(f) = [0; +\infty)$

$E(f) = [0; +\infty)$

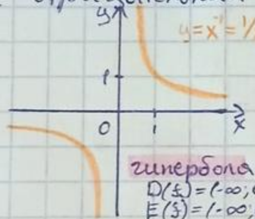
(4) α - нечетная дробная



$D(f) = \mathbb{R}$

$E(f) = \mathbb{R}$

(5) α - целая нечетная отрицательная

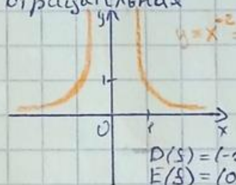


гипербола

$D(f) = (-\infty; 0) \cup (0; +\infty)$

$E(f) = (-\infty; 0) \cup (0; +\infty)$

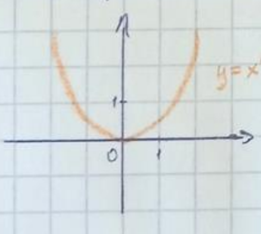
(6) α - целая четная отрицательная



$D(f) = (-\infty; 0) \cup (0; +\infty)$

$E(f) = (0; +\infty)$

• Квадратичная функция — $y = ax^2 + bx + c$, где $a \neq 0$



парабола

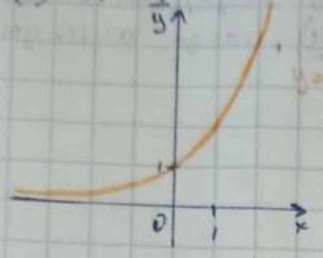
$D(f) = \mathbb{R}$

$E(f)$ = при $a > 0$ = $[-D/(4a); +\infty)$

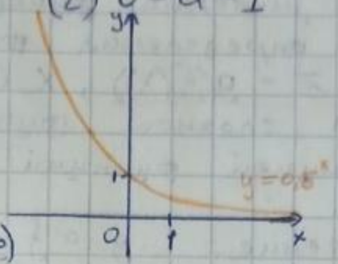
при $a < 0$ = $[-\infty; -D/(4a)]$

• показательная функция — $y = a^x$, $a > 0$

(1) $a > 1$



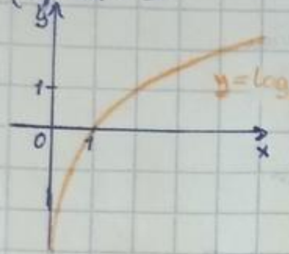
(2) $0 < a < 1$



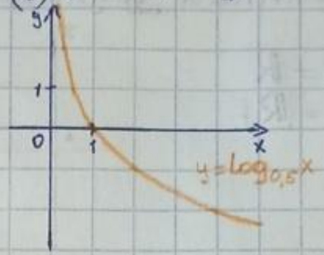
$D(f) = \mathbb{R}$
$E(f) = (0; +\infty)$

• логарифмическая функция — $y = \log_a x$, $a > 0$, $a \neq 1$

(1) $a > 1$



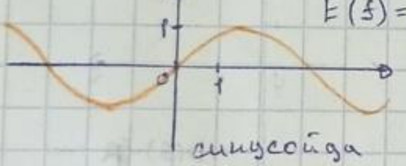
(2) $0 < a < 1$



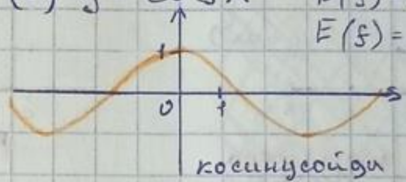
$D(f) = (0; +\infty)$
$E(f) = \mathbb{R}$

• тригонометрические функции:

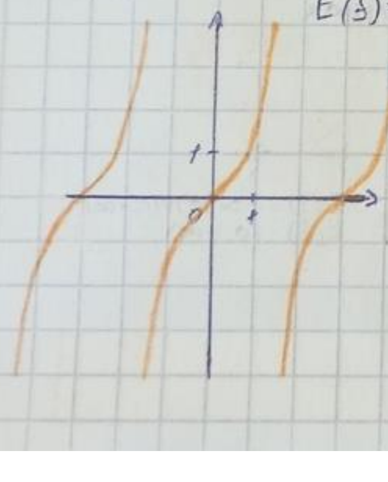
(1) $y = \sin x$ $D(f) = \mathbb{R}$
$E(f) = [-1; 1]$



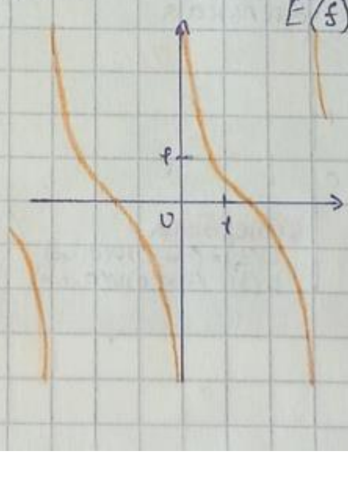
(2) $y = \cos x$ $D(f) = \mathbb{R}$
$E(f) = [-1; 1]$



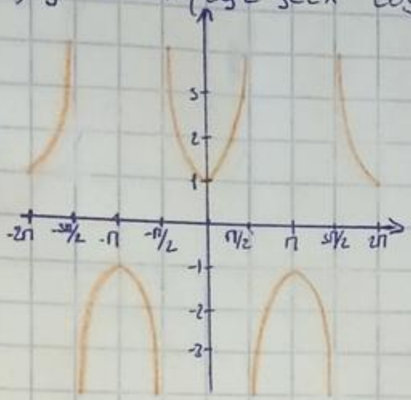
(3) $y = \operatorname{tg} x$ $D(f) =$ объедин. интервалов $(-\frac{\pi}{2} + \pi n; \frac{\pi}{2} + \pi n), n \in \mathbb{Z}$
$E(f) = \mathbb{R}$



(4) $y = \operatorname{ctg} x$ $D(f) =$ объедин. интервалов $(\pi n; \pi + \pi n), n \in \mathbb{Z}$
$E(f) = \mathbb{R}$



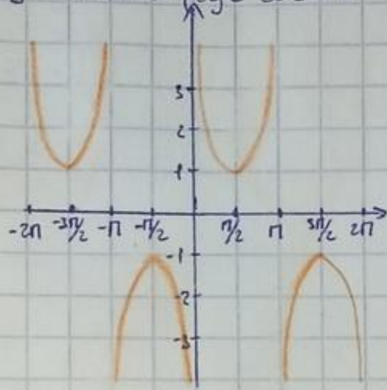
(5) $y = \sec x$ (где $\sec x = \frac{1}{\cos x}$)



$D(f) = \{x \mid x \neq \frac{\pi}{2} + \pi n\}$, $n \in \mathbb{Z}$

$E(f) = \{y \mid y \leq -1, y \geq 1\} = (-\infty; -1] \cup [1; +\infty)$

(6) $y = \operatorname{cosec} x$ (где $\operatorname{cosec} x = \frac{1}{\sin x}$)

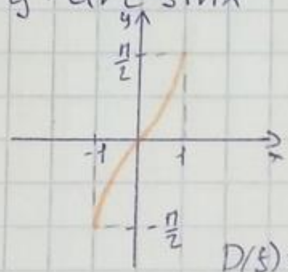


$D(f) = \{x \mid x \neq \pi n\}$, $n \in \mathbb{Z}$

$E(f) = \{y \mid y \leq -1, y \geq 1\} = (-\infty; -1] \cup [1; +\infty)$

• Обратные тригонометрические функции.

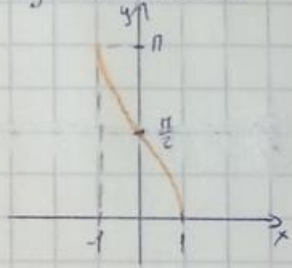
(1) $y = \arcsin x$



$D(f) = [-1; 1]$

$E(f) = [-\pi/2; \pi/2]$

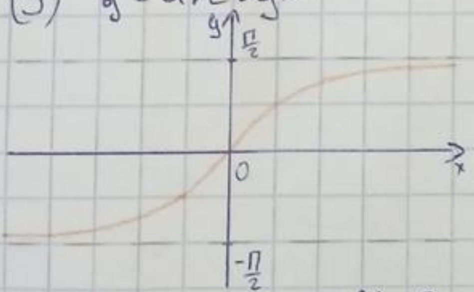
(2) $y = \arccos x$



$D(f) = [-1; 1]$

$E(f) = [0; \pi]$

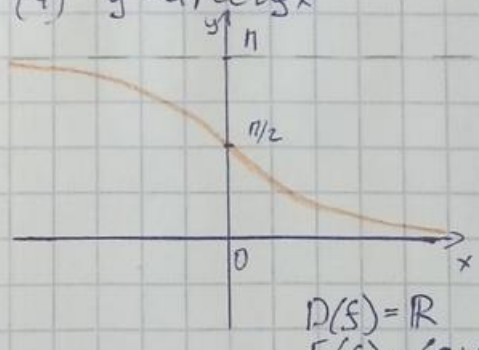
(3) $y = \operatorname{arctg} x$



$D(f) = \mathbb{R}$

$E(f) = (-\pi/2; \pi/2)$

(4) $y = \operatorname{arcctg} x$



$D(f) = \mathbb{R}$

$E(f) = (0; \pi)$